Title

Leonardo Luigi Pepe

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Abstract

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Contents

1 Introduction **2**
1.1 Preliminaries . . . 2
1.1.1 Graphs . . . 2
1.1.2 Matrices . . . 3

2 The French–Harary–DeGroot model **4**
2.1 Introduction to the FDG model . . . 4
2.1.1 The FDG model . . . 4
2.1.2 Consensus and social power . . . 4
2.1.3 Disagreement . . . 5
2.1.4 Rosetta stone . . . 6
2.2 Stubbornness and social power in the FDG model . . . 6
2.2.1 Modelling stubbornness: self-weights and stubborn agents . . . 7
2.2.2 Self-weights, stubbornness and convergence . . . 7
2.2.3 Self-weights, stubbornness and social power . . . 8

3 The Friedkin–Johnsen model **10**
3.1 Introduction to the FJ model . . . 10
3.1.1 The FJ model . . . 10
3.1.2 Convergence and stability . . . 10
3.1.3 How the opinions converge: the matrix V and the augmented graph . . . 12
3.1.4 Influence centrality . . . 14
3.2 Stubbornness and influence centrality in the FJ model . . . 15
3.2.1 Modelling stubbornness: prejudices and stubborn agents . . . 15
3.2.2 Prejudices, stubbornness and convergence . . . 19
3.2.3 Prejudices, stubbornness and influence centrality . . . 21

4 Conclusions **23**

1 Introduction

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides [2, 3, 1].

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \dots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides [1]; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction [5].

Definition 1.1 (Weighted adjacency matrix [2])**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n the (weighted) adjacency matrix of G is the $n \times n$ matrix A in which:

(i) for each edge $(i, j) \in E$ the (i, j) entry of A is equal to the edge weight a_{ij},

(ii) all other entries of A are equal to zero.

Definition 1.2 (Connectivity of digraphs [2])**.** A directed graph $G = (V, E)$ is

(i) *strongly connected* if there is a directed walk from any node to any other node;

(ii) *weakly connected* if the undirected version of the graph is connected.

Moreover, G has a

(iii) *globally reachable node* if one of its nodes can be reached from any other node through a directed walk;

(iv) *directed spanning tree* if one of its nodes is the root of directed walks to every other node.

With our convention, a directed walk from i to j means that agent i is influenced by agent j, either directly or through intermediaries. Hence G is strongly connected when every agent influences every other agent, at least indirectly, and a globally reachable node is an agent that influences the whole group.

Definition 1.3 (Periodic digraph [2])**.** A strongly connected digraph $G = (V, E)$ is *periodic* if there exists an integer $k > 1$ (the period) that divides the length of every simple cycle of the digraph.

A strongly connected digraph that is not periodic is called *aperiodic*; in particular, a strongly connected digraph with a self-loop is aperiodic, since the self-loop is a cycle of length 1.

Definition 1.4 (Strongly connected component [1])**.** Given a (weighted) digraph G. A subgraph S is a *strongly connected component* of G if S is strongly connected and any other subgraph of G strictly containing S is not strongly connected.

Definition 1.5 (Condensation digraph [1])**.** The *condensation digraph* $C(G)$ of a (weighted) digraph G is obtained by collapsing each strongly connected component of G into a single node. For any two distinct strongly connected components S_k and S_h there is an edge (S_k, S_h) in $C(G)$ iff there exists an edge (i, j) in G with $i \in S_k$ and $j \in S_h$.

A node with no outgoing edges is called a *sink* [2]. A sink of $C(G)$ is a group of agents that are influenced only by members of the group itself: no agent outside the group can change their opinions.

If G contains a globally reachable node, then $C(G)$ has a unique sink [1].

1.1.2 Matrices

Definition 1.6 (Non-negative and stochastic matrices [3]). A matrix $A = [a_{ij}] \in \mathbb{R}^{n\times n}$ is

(i) *non-negative* (*positive*) if $a_{ij} \geq 0$ ($a_{ij} > 0$, resp.) for all $i, j \in \{1, \ldots, n\}$;

(ii) *row-stochastic* if it is non-negative and $A\mathbf{1}_n = \mathbf{1}_n$;

(iii) *column-stochastic* if it is non-negative and $A^\top \mathbf{1}_n = \mathbf{1}_n$;

(iv) *doubly-stochastic* if it is row- and column-stochastic.

Theorem 1.7 (Spectral properties of row-stochastic matrices [3]). *For a row-stochastic matrix* $A \in \mathbb{R}^{n\times n}$

(i) $1 \in \operatorname{spec}(A)$*;*

(ii) $\operatorname{spec}(A)$ *is a subset of the unit disk and* $\rho(A) = 1$*.*

Definition 1.8 (Semi-convergent and convergent matrices [3]). A matrix $A \in \mathbb{R}^{n\times n}$ is

(i) *semi-convergent* if $\lim_{k\to+\infty} A^k$ exists (finite);

(ii) *convergent* (or *Schur stable*) if it is semi-convergent and $\lim_{k\to+\infty} A^k = 0_{n\times n}$.

Theorem 1.9 (Convergence and spectral radius [3]). *Consider a matrix* $A \in \mathbb{R}^{n\times n}$ *and let* $\rho(A) := \max\{|\lambda| \mid \lambda \in \operatorname{spec}(A)\}$ *(spectral radius of A). Then:*

(i) A convergent $\iff$ $\rho(A) < 1$*;*

(ii) A semi-convergent and not convergent $\iff$ 1 *is a semisimple eigenvalue of A and all the other eigenvalues have modulus strictly less than* 1*.*

Definition 1.10 (Regularity [5]). We call the matrix A *regular* if the limit $A^\infty = \lim_{k\to\infty} A^k$ exists and *fully regular* if, additionally, the rows of A^∞ are identical (that is, $A^\infty = \mathbf{1}_n w^\top$ for some $w \in \mathbb{R}^n$).

In the terminology of the course slides, a matrix is regular if and only if it is semi-convergent (Definition 1.8).

Definition 1.11 (Irreducible and primitive matrices [3]). A non-negative matrix $A \in \mathbb{R}^{n\times n}$, $n \geq 2$, is

(i) *irreducible* if $\sum_{k=0}^{n-1} A^k$ is a positive matrix;

(ii) *primitive* if there exists $k \in \mathbb{N}$ such that A^k is positive;

(iii) *reducible* if it is not irreducible.

The next theorem uses two further notions [2]: $\{I, J\}$ is a *partition* of $V = \{1, \ldots, n\}$ if $I, J \neq \emptyset$, $I \cap J = \emptyset$ and $I \cup J = V$; $P \in \{0, 1\}^{n\times n}$ is a *permutation matrix* if it has exactly one entry equal to 1 in every row and every column.

Theorem 1.12 (Irreducible matrices and digraphs [2]). *Let* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ *be a weighted digraph of order* $n \geq 2$ *with adjacency matrix A. The following statements are equivalent:*

(i) A is irreducible;

(ii) there exists no permutation matrix P such that $PAP^\top$ *is block upper triangular;*

(iii) G is strongly connected;

(iv) for all partitions $\{I, J\}$ *of* V*, there exists* $i \in I$ *and* $j \in J$ *such that* $(i, j) \in E$*.*

Theorem 1.13 (Primitive matrices and digraphs [2]). *Let* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ *be a weighted digraph of order* $n \geq 2$ *with adjacency matrix A. The following statements are equivalent:*

(i) A is primitive,

(ii) G is strongly connected and aperiodic.

Corollary 1.14 (Primitive matrices and digraphs with self-loops [2])**.** *Let $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ be a weighted digraph of order $n \geq 2$ with adjacency matrix A. If G has at least one self-loop and is strongly connected, then A is primitive.*

Given a weighted digraph G with weighted adjacency matrix A, the *weighted out-degree* of node i is the i-th row sum $\sum_{j=1}^{n} a_{ij}$ of A [2]. A row-stochastic matrix has all weighted out-degrees equal to 1; when some of them are smaller than 1, the following notions apply.

Definition 1.15 (Row-substochastic matrix [1])**.** A non-negative matrix $A \in \mathbb{R}^{n\times n}$ is *row-substochastic* if its row-sums are at most 1 and at least one row-sum is strictly less than 1. That is:

$$A\mathbf{1}_n \leq \mathbf{1}_n \quad \text{and} \quad \exists\, i \in \{1, \dots, n\} \text{ s.t. } e_i^\top A\mathbf{1}_n < 1.$$

Lemma 1.16 (Convergent row-substochastic matrices [1])**.** *Let G be a weighted digraph with row-substochastic adjacency matrix A. Then, A is convergent if and only if G contains directed walks from each node with (weighted) out-degree 1 to a node with (weighted) out-degree strictly less than 1.*

2 The French–Harary–DeGroot model

2.1 Introduction to the FDG model

2.1.1 The FDG model

Consider a group of agents $V = \{1, 2, \dots, n\}$, where $x_i \in \mathbb{R}$ is the opinion of agent i on a given topic, and a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ describing who influences whom, with the convention of Section 1.1. In the French–Harary–DeGroot (FDG) model [1], called French–DeGroot model in [5], the new opinion of each agent is a weighted average of the opinions of the agents it is influenced by,

$$x_i(k+1) = \sum_{j=1}^{n} a_{ij}\, x_j(k), \qquad \sum_{j=1}^{n} a_{ij} = 1, \tag{1}$$

or, in vector form, $x(k+1) = Ax(k)$, where A is the weighted adjacency matrix of G (Definition 1.1). The constraint on the weights means that A is row-stochastic (Definition 1.6). In the terminology of [1], the FDG model is a *DT averaging system*.

Since $x(k) = A^k x(0)$, the asymptotic behaviour of the model is determined by the powers of A. Three outcomes are possible:

- *no convergence*: $\lim_{k\to+\infty} x(k)$ does not exist for some $x(0)$; this happens when A is not semi-convergent (Definition 1.8), e.g. when A has an eigenvalue $\mu \neq 1$ with $|\mu| = 1$;
- *consensus*: for every $x(0)$, $\lim_{k\to+\infty} x(k) = \alpha\mathbf{1}_n$ for some $\alpha \in \mathbb{R}$, which depends on $x(0)$ [5];
- *disagreement*: $\lim_{k\to+\infty} x(k) = v$ with $v \neq \alpha\mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

2.1.2 Consensus and social power

The following theorem characterizes when the FDG model reaches a consensus. Its condition (A3) involves the subgraph of globally reachable nodes, which is always strongly connected [1], so that its periodicity is well defined (Definition 1.3).

Theorem 2.1 (Consensus in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. The following statements are equivalent:*

(A1) the eigenvalue 1 is simple and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent and $\lim_{k\to\infty} A^k = \mathbf{1}_n w^\top$ where $w \in \mathbb{R}^n$ satisfies $w \geq 0$, $\mathbf{1}_n^\top w = 1$, and $w^\top A = w^\top$;

(A3) G contains a globally reachable node and the subgraph of globally reachable nodes is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then

(i) w is the left dominant eigenvector of A and $w_i > 0$ iff node i is globally reachable;

(ii) the solution of the DT averaging system satisfies $\lim_{k\to\infty} x(k) = \big(w^\top x(0)\big)\mathbf{1}_n$.

Corollary 2.2 (Average consensus [1])**.** *If A is doubly-stochastic and (A3) holds, then $w = \frac{1}{n}\mathbf{1}_n$ and $\lim_{k\to+\infty} x(k) = \big(\frac{1}{n}\sum_{i=1}^n x_i(0)\big)\,\mathbf{1}_n$.*

In Theorem 2.1, the left dominant eigenvector is the left eigenvector associated with the dominant eigenvalue $\rho(A) = 1$ (Theorem 1.7) [3].

Social power. The entry w_i is called the *social power* of agent i: it is the weight of the initial opinion $x_i(0)$ in the final opinion of the group, i.e. in the consensus value [5]

$$\lim_{k\to\infty} x_i(k) = w^\top x(0) = \sum_{j=1}^n w_j\, x_j(0) \qquad \text{for every agent } i. \tag{2}$$

This definition requires a consensus: by (A2), all the rows of $\lim_{k\to\infty} A^k = \mathbf{1}_n w^\top$ are equal to $w^\top$, so a single vector describes the whole group. Without a consensus, as in Theorem 2.3, the rows of the limit matrix differ, each agent ends at its own weighted average of the initial opinions, and no single vector plays the role of w.

Point (i) has a direct social interpretation: the consensus value is the same for every agent, so the initial opinion of agent i can enter it only if it reaches *all* the agents, i.e. if i is globally reachable. Hence the consensus value depends only on the initial opinions of the globally reachable agents.

Among the globally reachable agents, the social power depends on their position in the network, and it can be regarded as a centrality measure [5]. For instance, in the model of French with $n = 3$ agents and

$$A = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/3 & 1/3 & 1/3 \\ 0 & 1/2 & 1/2 \end{bmatrix},$$

all agents are globally reachable, but agent 2, the only one listened to by all the others, has the largest social power, $w = (2/7,\, 3/7,\, 2/7)^\top$ [5].

2.1.3 Disagreement

Theorem 2.1 requires a single globally reachable group, i.e. a unique sink in the condensation digraph $C(G)$. We now consider the case in which $C(G)$ has $n_s \geq 2$ sinks. Each sink is a strongly connected component, so its periodicity is well defined (Definition 1.3).

Theorem 2.3 (Disagreement in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. Assume that the condensation digraph $C(G)$ has $n_s \geq 2$ sinks. The following statements are equivalent:*

(A1) the eigenvalue 1 is semi-simple with multiplicity n_s and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent;

(A3) each sink of $C(G)$ is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then the solution of the DT averaging system satisfies

$$\lim_{k\to\infty} x(k) = v \neq \alpha\mathbf{1}_n, \quad \alpha \in \mathbb{R}.$$

The final opinions can be described group by group. The agents of a sink are not influenced by anyone outside it, so each sink evolves as an FDG model on its own and, being aperiodic, reaches an internal consensus that depends only on the initial opinions of its members; every agent outside the sinks converges to a convex combination of the consensus values of the sinks [1]. In social terms, a sink is a closed group that listens only to itself, and the agents in between settle on an intermediate opinion: the disagreement is bounded by the smallest and the largest consensus value of the sinks.

2.1.4 Rosetta stone

Table 1, the so-called *Rosetta stone* [1], summarizes all the possible behaviours of the FDG model: it translates each property of the row-stochastic matrix A into a property of the digraph G and into the asymptotic outcome, with an example for each case; the translation between A and G comes from Theorems 1.12 and 1.13 in cases (i)–(ii), and from Theorems 2.1 and 2.3 in cases (iii)–(iv). In case (i) both agents are globally reachable and A is doubly-stochastic, so by Corollary 2.2 the group agrees on the average of the initial opinions. In case (iii) agent 1 is stubborn (Definition 2.4) and is the only globally reachable agent: $w = (1, 0)^\top$ and the group agrees on $x_1(0)$. In case (iv) both agents are stubborn: each of them is an aperiodic sink of $C(G)$, and every agent keeps its initial opinion (Theorem 2.3).

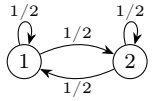
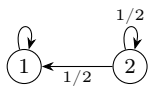
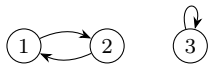
Case	Example digraph	A	spec(A)	Property of A / of G	Outcome
(i)		$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$	{1, 0}	A primitive / G strongly connected and aperiodic	consensus
(ii)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	{1, −1}	A irreducible, not primitive / G strongly connected and periodic	no convergence
(iii)		$\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$	{1, 1/2}	1 simple and strictly dominant, A not primitive / $C(G)$ has a unique sink, which is aperiodic	consensus
(iv)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	{1, 1}	1 semi-simple but not simple, strictly dominant / $C(G)$ has at least two sinks, all aperiodic	disagreement
(v)		$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	{1, 1, −1}	1 not strictly dominant / G not strongly connected, at least one sink of $C(G)$ is periodic	no convergence

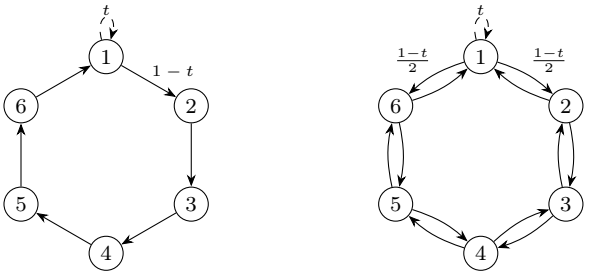
Table 1: Rosetta stone of the FDG model $x(k + 1) = Ax(k)$, with A row-stochastic: for each case, an example digraph (unlabelled edges have weight 1), its matrix A, the spectrum of A, the general property of A and of G, and the outcome. Cases (i)–(iv) and the examples follow [1]; the properties of case (v), left blank in the slides, follow from [5].

2.2 Stubbornness and social power in the FDG model

In the FDG model the simplest way to account for stubbornness is the self-weight a_{ii}, i.e. the weight of the self-loop of agent i. This section studies how self-weights, and stubborn agents in particular, affect the two aspects introduced in Section 2.1: whether the group reaches an agreement, and how the social power is distributed among the agents.

All the examples use the two networks of Figure 1, both with $n = 6$ agents: in the *directed cycle* each agent is influenced only by the next one, $a_{i,i+1} = 1$; in the *bidirectional ring* by its two neighbours, $a_{i,i-1} = a_{i,i+1} = 1/2$ (indices modulo 6). Socially, in the first network everyone listens to a single person, in the second to the two people next to them. Without self-loops both digraphs are strongly

connected and periodic, with period 6 for the directed cycle and 2 for the bidirectional ring, whose cycles all have even length: neither network reaches an agreement (case (ii) of Table 1). Both matrices are doubly-stochastic, so $w = \frac{1}{6}\mathbf{1}_6$ is the left eigenvector associated with the eigenvalue 1: no agent is favoured by the structure of the network, and any asymmetry in the social power will be due only to the self-weight added to agent 1.



(a) directed cycle (b) bidirectional ring

Figure 1: The two networks used in Sections 2.2 and 3. An edge $i \to j$ means that agent i is influenced by agent j. In (a) all edges have weight 1, in (b) all edges have weight 1/2. The dashed self-loop of agent 1, of weight t, is the only modification considered: the other edges of agent 1 are scaled by $1-t$, as marked (equation (3)).

2.2.1 Modelling stubbornness: self-weights and stubborn agents

The self-weight a_{ii} is the weight that agent i gives to its own *current* opinion [4]; it indicates how open the agent is to the opinions of the others [5]. An agent with $a_{ii} = 0$ is open-minded and relies completely on the others; an agent with a positive self-weight changes its opinion more slowly, but it still moves towards the opinions it listens to. In social terms, a self-weight is a form of inertia, not an anchor to a fixed opinion.

Definition 2.4 (Stubbornness [5])**.** An agent is said to be *stubborn* if its opinion remains unchanged independent of the others' opinions.

If the opinions obey the FDG model, then agent i is stubborn, $x_i(k) \equiv x_i(0)$, if and only if $a_{ii} = 1$ [5] (and hence $a_{ij} = 0$ for all $j \neq i$, since A is row-stochastic).

Stubbornness is therefore the extreme case of the self-weight, and there is no intermediate degree of it: an agent with $0 < a_{ii} < 1$ is not stubborn, it only has a positive self-weight.

In the digraph, a stubborn agent has no edges towards the other agents: it is influenced by nobody, while the agents that listen to it keep doing so. Hence it forms a strongly connected component made of a single node, which is a sink of $C(G)$ and is aperiodic because of its self-loop. If, in addition, every agent is influenced by the stubborn agent i, directly or through intermediaries, then i is the only globally reachable node, and Theorem 2.1 gives $w = e_i$: the whole group converges to the initial opinion of the stubborn agent (see also [5]).

To move continuously from no self-weight to stubbornness, we modify only the row of agent 1. Given a row-stochastic matrix A with $a_{11} = 0$, as in both networks of Figure 1, we replace its first row by

$$a_1(t)^\top = (1-t)\, a_1^\top + t\, e_1^\top, \qquad t \in [0,1], \tag{3}$$

where e_1 is the first vector of the canonical basis, leaving the other rows unchanged: agent 1 gives weight t to itself and splits the remaining weight $1-t$ among the agents it listened to before, in the same proportions, as marked in Figure 1. Thus $t = a_{11}$: $t = 0$ is the original network and $t = 1$ makes agent 1 stubborn. In the following we consider $t = 0$, $1/2$ and 1.

2.2.2 Self-weights, stubbornness and convergence

The three values of t correspond to three different cases of the Rosetta stone.

- $t = 0$: both digraphs are strongly connected and periodic, so neither network converges (case (ii)).

- $0 < t < 1$: agent 1 still listens to its neighbours with total weight $1 - t > 0$, so the digraph stays strongly connected, and the self-loop of agent 1 is a cycle of length 1, so the digraph becomes aperiodic. By Corollary 1.14 the matrix is primitive, and the group reaches a consensus (case (i)).
- $t = 1$: agent 1 is stubborn and influences every other agent through the cycle, so it is the only globally reachable node (Section 2.2.1) and the group reaches a consensus on $x_1(0)$ (case (iii)).

The same picture can be read on the spectrum of $A(t)$. For $t = 0$ the spectrum is the set of the sixth roots of unity for the directed cycle and $\{1, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -1\}$ for the bidirectional ring: in both cases some eigenvalues different from 1 lie on the unit circle, and they cause the oscillations. For $t > 0$ all the eigenvalues different from 1 lie strictly inside the unit disk, and the group converges; for $t = 1/2$ the largest of their moduli is ≈ 0.92 on the directed cycle and ≈ 0.90 on the bidirectional ring. For $t = 1$ the directed cycle is left with the eigenvalue 0, of multiplicity 5, which is why its consensus is reached in a finite number of steps.

Figure 2 shows the opinions $x_i(k)$ in the six cases. For $t = 0$ the opinions keep rotating along the directed cycle and oscillating between two configurations on the bidirectional ring. For $t = 1/2$ both networks reach a consensus on a weighted average of all the initial opinions (Section 2.2.3). For $t = 1$ both reach $x_1(0)$, in different ways: in the directed cycle the opinion of agent 1 is copied backwards along the chain, one agent per step, and the consensus is reached after $n - 1 = 5$ steps; in the bidirectional ring the opinions approach $x_1(0)$ gradually, the neighbours of agent 1 first.

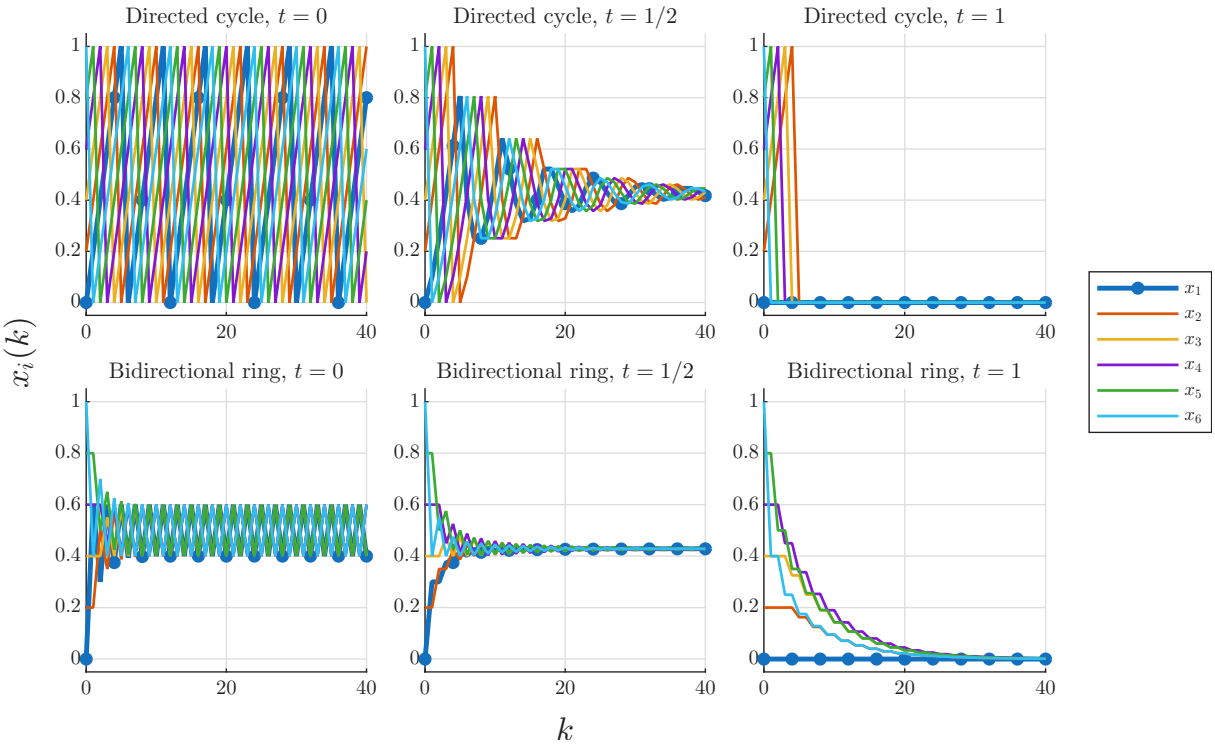


Figure 2: Opinions $x_i(k)$ in the two networks of Figure 1 (top: directed cycle; bottom: bidirectional ring) for three values of the self-weight t of agent 1 (columns), with $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$. Agent 1 is marked with dots.

In social terms, a group in which nobody gives weight to its own opinion can keep passing opinions around forever; a single agent that gives some weight to its current opinion, without being stubborn, is enough for the whole group to agree, and if that agent becomes stubborn the agreement is on its own opinion. The network decides only how this happens: by copying along a chain, or by averaging with the neighbours.

2.2.3 Self-weights, stubbornness and social power

For $t > 0$ both networks reach a consensus (Section 2.2.2), so the social power w of Section 2.1.2 is well defined, and Theorem 2.1 describes how it depends on t.

- $0 < t < 1$: the digraph is strongly connected, so every agent is globally reachable and, by point (i) of Theorem 2.1, every agent has a positive social power. Agent 1 gives weight to its own opinion, but it does not take all the influence: the initial opinion of every agent enters the consensus value.

- $t = 1$: agent 1 is stubborn and is the only globally reachable node, so $w = e_1$: the stubborn agent takes all the social power, and the initial opinions of the other agents have no effect on the final opinion of the group.

Between these two extremes, w is the left eigenvector of $A(t)$ associated with the eigenvalue 1, normalized to sum 1 (condition (A2) of Theorem 2.1); Figure 3 reports it for $t \in (0, 1]$. Numerically, the social power of agent 1 grows with t towards 1, while the other agents share the rest equally ($w_1 \approx 0.286$ and $w_i \approx 0.143$ for $i \neq 1$ when $t = 1/2$), with the same values on the two networks. The bottom panel shows where the group converges, the consensus value $w^\top x(0)$ (point (ii) of Theorem 2.1): it moves from the average 0.5 of the initial opinions, for small t, to $x_1(0) = 0$ for $t = 1$, and it is ≈ 0.429 for $t = 1/2$, the value reached in Figure 2; also this curve is the same on the two networks. In other words, on these two networks the self-weight of agent 1 changes how the opinions converge, which depends on the network, but not where they converge.

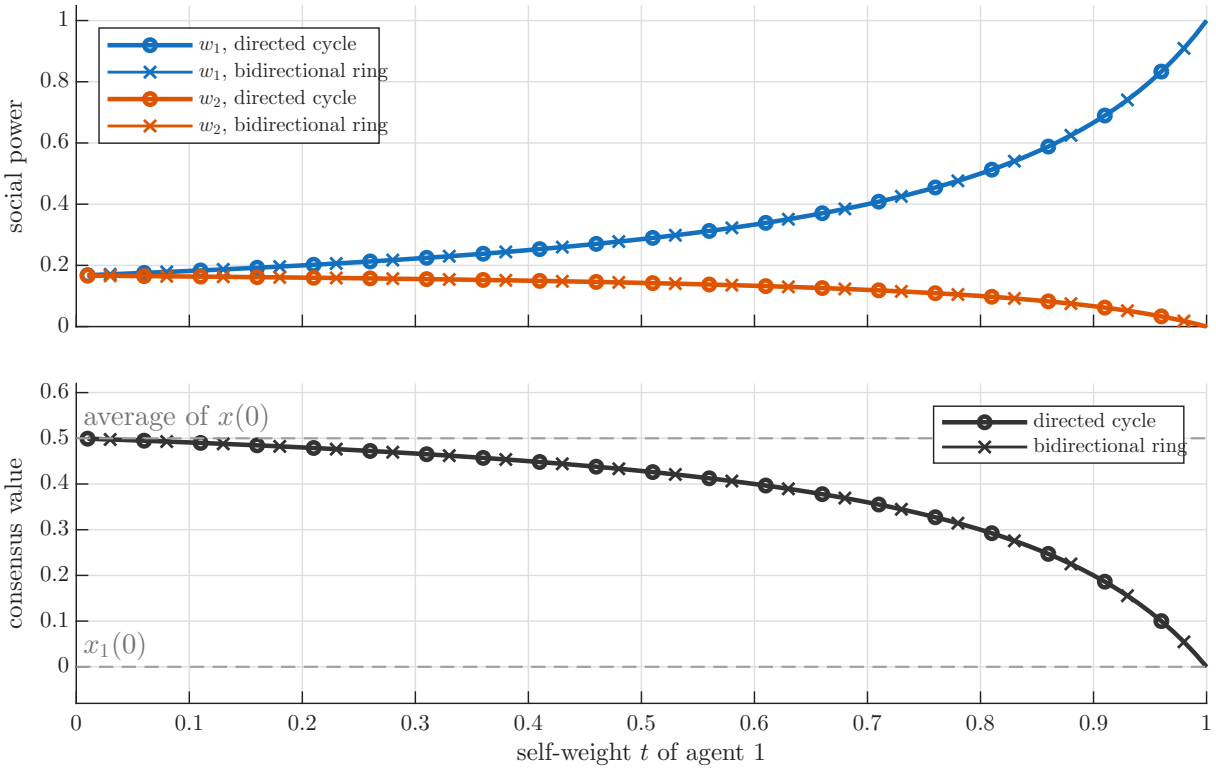


Figure 3: Top: social power of agent 1 (w_1) and of agent 2 (w_2) as a function of the self-weight $t \in (0, 1]$ of agent 1. Bottom: consensus value $w^\top x(0)$ for $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$; the dashed lines mark the average of $x(0)$ and $x_1(0)$. Both panels are computed numerically in the directed cycle (dots) and in the bidirectional ring (crosses); the two networks give the same values.

Remark 2.5 (More than one stubborn agent). If two or more agents are stubborn, each of them is a sink of $C(G)$ made of a single, aperiodic node. If every other agent is influenced, directly or through intermediaries, by at least one stubborn agent, these are the only sinks, and by Theorem 2.3 the group converges, but to a disagreement: every stubborn agent keeps its own initial opinion, and every other agent converges to a convex combination of the opinions of the stubborn agents it is influenced by (see also [5]). Since there is no single consensus value, the social power is no longer defined. This situation, in which several agents are anchored to different fixed opinions, is the starting point of the Friedkin–Johnsen model of Section 3.

3 The Friedkin–Johnsen model

3.1 Introduction to the FJ model

3.1.1 The FJ model

In the FDG model every agent keeps averaging the opinions it listens to, and a persistent disagreement can only come from the structure of the network, i.e. from several closed groups (Theorem 2.3). The Friedkin–Johnsen (FJ) model adds a second ingredient: each agent can also be tied to a fixed opinion of its own, called its *prejudice*. Following [5], the model reads

$$x(k+1) = \Lambda W x(k) + (I - \Lambda)\, u, \tag{4}$$

where

- $W \in \mathbb{R}^{n\times n}$ is a row-stochastic matrix, the *matrix of social influences*, i.e. the weighted adjacency matrix of the influence digraph G (Definition 1.1), which plays the role of A in the FDG model;
- $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$ the *susceptibility* of agent i to social influence;
- $u \in \mathbb{R}^n$ is a constant vector collecting the agents' *prejudices*.

Componentwise,

$$x_i(k+1) = \lambda_i \sum_{j=1}^{n} w_{ij}\, x_j(k) + (1 - \lambda_i)\, u_i. \tag{5}$$

At every step agent i averages the opinions it listens to, exactly as in the FDG model, and mixes this average with its prejudice: the weight λ_i goes to the others and the complement $1 - \lambda_i$ goes to u_i. The two quantities play different roles: the prejudice u_i is an opinion and says *where* the agent is anchored, while $1 - \lambda_i$ is a weight and says *how strongly*. The model with $u = x(0)$ appears in [4], where λ_i is called the openness level of agent i and $1 - \lambda_i$ its attachment to its initial opinion, which plays the role of the prejudice.

The special cases of (4) are the following [5].

- If $\lambda_i = 1$, agent i follows the FDG rule. If $\Lambda = I$, the FJ model reduces to the FDG model with $A = W$; for this reason we assume $\Lambda \neq I$ from now on.
- If $\lambda_i < 1$, agent i is called *prejudiced*: it is anchored to its prejudice u_i.
- If $\lambda_i = 0$, then $x_i(k) = u_i$ for all $k \geq 1$: the agent ignores the others and jumps to its prejudice after one step. Its opinion never changes, i.e. the agent is stubborn in the sense of Definition 2.4, only if in addition $u_i = x_i(0)$. A stubborn agent of the FJ model is therefore an agent with $\lambda_i = 0$ and $u_i = x_i(0)$.

In social terms, the self-weight of the FDG model and the prejudice of the FJ model are two different forms of resistance to influence. A self-weight ties an agent to its *current* opinion, which still moves towards the others (Section 2.2.1); a prejudice ties it to a *fixed* opinion, which never moves.

In the original formulation of Friedkin and Johnsen the prejudices coincide with the initial opinions, $u = x(0)$, and the susceptibilities are coupled to the self-weights through $1 - \lambda_i = w_{ii}$ [5, 4]. As in [5], we do not impose these conditions: u and $x(0)$ are independent, and so are Λ and W. The choice $u = x(0)$ will be needed in Section 3.1.4 to define the influence centrality.

3.1.2 Convergence and stability

The FJ model (4) is a linear system with the constant input u. Following [5], we say that it *converges* if the limit $x(\infty) = \lim_{k\to+\infty} x(k)$ exists for every initial condition $x(0)$, and that it is *asymptotically stable* if ΛW is convergent, i.e. $\rho(\Lambda W) < 1$ (Definition 1.8 and Theorem 1.9). Regular matrices are recalled in Definition 1.10.

Both properties are characterized by distinguishing the agents that are reached by the prejudices from those that are not.

Definition 3.1 (P-dependent agents [5])**.** Agent i is *P-dependent* if it is prejudiced or influenced by some prejudiced agent j, that is, a walk from i to j in the graph G exists. Otherwise, the agent is *P-independent.*

The walk goes from the agent to the prejudiced agent because of our convention on the edges; in [5], where the edges are drawn in the opposite direction, the same walk goes from the prejudiced agent to the agent. In social terms, an agent is P-dependent if at least one prejudice reaches it.

A P-independent agent i is not prejudiced, so $\lambda_i = 1$, and it is not influenced by any P-dependent agent: if $w_{ij} > 0$ with j P-dependent, there would be a walk from i to a prejudiced agent through j, and i would be P-dependent. Since $\Lambda \neq I$, there is at least one prejudiced agent, hence at least one P-dependent agent. Renumbering the agents so that agents $1, \dots, r$, with $r \geq 1$, are P-dependent and agents $r+1, \dots, n$ are P-independent, and splitting $x = [x^1; x^2]$, $u = [u^1; u^2]$, W and Λ accordingly, we have $W^{21} = 0$ and $\Lambda^{22} = I$, and the model (4) decomposes as [5]

$$x^1(k+1) = \Lambda^{11}\left[W^{11}x^1(k) + W^{12}x^2(k)\right] + (I - \Lambda^{11})\,u^1, \tag{6}$$

$$x^2(k+1) = W^{22}x^2(k), \tag{7}$$

where W^{22} is row-stochastic, since W is row-stochastic and $W^{21} = 0$. The P-independent agents therefore follow an FDG model among themselves, with $A = W^{22}$, and no prejudice reaches them; the P-dependent agents receive the opinions of the P-independent ones as an additional input.

Theorem 3.2 ([5])**.** *Let the community have $r \geq 1$ P-dependent agents and $n - r \geq 0$ P-independent ones. Then, the subsystem* (6) *is asymptotically stable, i.e. $\Lambda^{11}W^{11}$ is Schur stable $\rho(\Lambda^{11}W^{11}) < 1$. The model* (4) *is convergent if and only if $r = n$ or* (7) *is convergent, i.e. W^{22} is regular. In this case*

$$x^1(\infty) = V\begin{bmatrix} u^1 \\ x^2(\infty) \end{bmatrix}, \qquad V \triangleq (I_r - \Lambda^{11}W^{11})^{-1}\begin{bmatrix} I_r - \Lambda^{11} & \Lambda^{11}W^{12} \end{bmatrix}. \tag{8}$$

The matrix V is stochastic, i.e. the final opinion of any agent is a convex combination of the prejudices and final opinions of P-independent agents.

In [5], *stochastic* stands for row-stochastic in the sense of Definition 1.6.
Theorem 3.2 can be read according to the number r of P-dependent agents.

- $r = 0$. This case is not covered by Theorem 3.2: no agent is prejudiced, i.e. $\Lambda = I$, and the FJ model (4) reduces to the FDG model with $A = W$ (Section 2.1). This is why we assume $\Lambda \neq I$, which guarantees $r \geq 1$.
- $0 < r < n$. P-dependent and P-independent agents coexist. The P-independent agents follow (7), which is an FDG model with the row-stochastic matrix W^{22}, so the outcomes of Table 1 apply to them. Denoting by C^{22} the condensation digraph of the subgraph of P-independent agents, whose adjacency matrix is W^{22}, they reach
  - a consensus, if C^{22} has a unique sink, which is aperiodic (cases (i) and (iii));
  - a disagreement, if C^{22} has several sinks, all aperiodic (case (iv));
  - no convergence, if at least one sink of C^{22} is periodic (cases (ii) and (v)).

  In the first two cases W^{22} is regular, the model (4) converges and the final opinions of the P-dependent agents are given by (8); in the third case the model (4) does not converge.
- $r = n$. There are no P-independent agents, the subsystem (6) is the whole model and the model (4) always converges. This case is the subject of Corollary 3.3 below.

In social terms, the P-dependent agents settle on a compromise between the prejudices and the opinions of the P-independent agents they listen to whenever these opinions settle; whether the whole group settles is decided only by the P-independent agents.

Corollary 3.3 ([5])**.** *The Friedkin–Johnsen model* (4) *is asymptotically stable if and only if all agents are P-dependent. Then, $V = (I - \Lambda W)^{-1}(I - \Lambda)$. This holds, in particular, if $\Lambda < I_n$ or $\Lambda \neq I_n$ and G is strongly connected (W is irreducible).*

With $r = n$, (8) gives, for every $x(0)$,

$$x(\infty) = Vu, \tag{9}$$

with V as in Corollary 3.3.

The condition of Corollary 3.3 can also be read on the matrix ΛW with the tools of the course. The matrix ΛW is no longer row-stochastic: since W is row-stochastic, the i-th row of ΛW sums to λ_i. Rows of agents that are not prejudiced sum to 1, rows of prejudiced agents sum to less than 1, and since $\Lambda \neq I$ at least one such row exists. Hence ΛW is row-substochastic (Definition 1.15) and Lemma 1.16 applies.

For the matrix ΛW, the weighted out-degree of agent i is λ_i: the nodes with out-degree 1 are the agents that are not prejudiced, and the nodes with out-degree less than 1 are the prejudiced agents. The condition of Lemma 1.16 therefore reads: every agent that is not prejudiced is influenced, directly or through intermediaries, by a prejudiced agent, i.e. all the agents are P-dependent. This is exactly the condition of Corollary 3.3. Indeed, [5] remarks that Corollary 3.3 can be turned into a criterion of Schur stability for substochastic matrices, and [4] proves the same condition for the FJ model through this lemma.

Two observations of ours give an intuition for Corollary 3.3. First, letting $k \to +\infty$ in (4) shows that the limit $x(\infty) = Vu$ is an equilibrium, $x(\infty) = \Lambda W x(\infty) + (I - \Lambda)u$. Letting $e(k) = x(k) - x(\infty)$ and subtracting this equation from (4) gives

$$e(k+1) = \Lambda W\, e(k), \tag{10}$$

an autonomous linear system in which the prejudice u does not appear: ΛW decides whether the opinions converge, u decides where.

Second, the linear system (10) shows the role of the P-dependent agents. At each step, agent i keeps a fraction λ_i of the weighted average of the deviations of the agents it listens to. An agent that is not prejudiced ($\lambda_i = 1$) only averages the deviations, while a prejudiced agent loses a fraction $1 - \lambda_i$ of them (in (4), this fraction is replaced by the agent's fixed prejudice). The deviations therefore leave the group only through the prejudiced agents, and they vanish only if every agent is influenced, directly or through intermediaries, by one of them, i.e. all agents are P-dependent. A group of agents that is not influenced, even indirectly, by any prejudiced agent loses no deviation and behaves as an FDG model, as in (7).

In social terms, the group converges, for every initial opinion, as soon as every agent that is not anchored to a prejudice listens, directly or through intermediaries, to someone who is. On a strongly connected network, such as the two networks of Figure 1, a single prejudiced agent is enough. Note that this does not require aperiodicity, unlike the FDG model of Section 2.2.

If some agents are P-independent, the model is not asymptotically stable, but it may still converge. Theorem 3.2 gives the condition in terms of W^{22}; in terms of the whole matrix ΛW it reads as follows.

Corollary 3.4 ([5]). *The model* (4) *converges if and only if* ΛW *is regular, i.e. the limit* $\lim_{k\to\infty}(\Lambda W)^k$ *exists.*

As remarked in [5], this property is not trivial: a system $x(k+1) = Ax(k) + Bu$ with a regular matrix A may not converge, and may even have unbounded solutions, as for $A = B = I$.

3.1.3 How the opinions converge: the matrix V and the augmented graph

When all agents are P-dependent, the final opinions are given by (9), and the convex combination of Theorem 3.2 involves only the prejudices:

$$x_i(\infty) = \sum_{j=1}^{n} v_{ij}\, u_j.$$

Three facts follow.

- The initial opinions are forgotten: $x(0)$ does not appear in $x(\infty)$.

- Only the prejudices of prejudiced agents count: if $\lambda_j = 1$, the j-th column of $I - \Lambda$ is zero, hence so is the j-th column of V, and u_j has no weight in any final opinion.
- Every final opinion lies between the smallest and the largest prejudice of the prejudiced agents.

The comparison with the FDG model is instructive. There, the consensus value $w^\top x(0)$ is a weighted average of the initial opinions, with the same weights w for every agent (Theorem 2.1); here, the final opinions are weighted averages of the prejudices, and each agent has its own weights, given by its row of V.

In the FDG model, consensus is a property of the model: the model reaches a consensus if the final opinions coincide for any initial condition $x(0)$ [5]. In the FJ model with all agents P-dependent, the limit $x(\infty) = Vu$ does not depend on $x(0)$ (equation (9)): whether the final opinions coincide is decided by the prejudices u, which are an external input, through the matrix V, and no longer by the initial opinions. In our reading, this is why [5] studies the FJ model in terms of convergence and of the form of the limit, and not in terms of consensus.

The notion of consensus of the FDG model applies again only to the P-independent agents ($0 < r < n$), which evolve as an FDG model with matrix W^{22} and whose limit depends on their initial opinions $x^2(0)$ (Section 3.1.2). The P-dependent agents converge to convex combinations of the prejudices and of the final opinions of the P-independent agents (Theorem 3.2), so they do not necessarily end up at the consensus value of the P-independent agents, even when there is one.

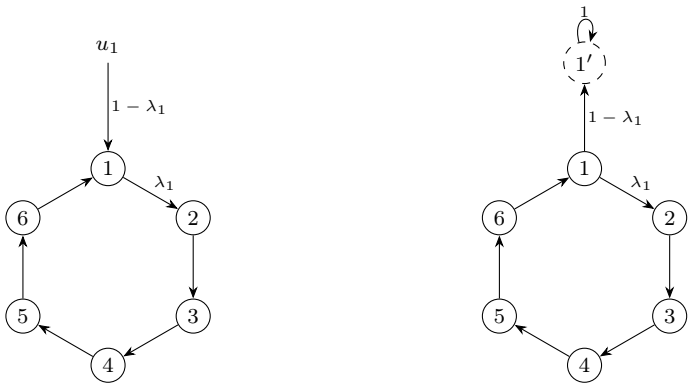
The limit (9) has a direct interpretation in terms of the FDG model. Following [5], the FJ model (4) may be considered as an augmented FDG model with n virtual stubborn agents, anchored at the prejudices: the virtual agent i' associated with agent i keeps the opinion $x_{i'}(k) = u_i$ for all k. Explicitly (this rewriting is ours), stacking the opinions x of the real agents and x' of the virtual agents,

$$\begin{bmatrix} x(k+1) \\ x'(k+1) \end{bmatrix} = \begin{bmatrix} \Lambda W & I - \Lambda \\ 0 & I \end{bmatrix} \begin{bmatrix} x(k) \\ x'(k) \end{bmatrix}, \qquad x'(0) = u. \tag{11}$$

Row i of the matrix in (11) sums to $\lambda_i + (1 - \lambda_i) = 1$, and row i' sums to 1; hence the matrix is row-stochastic and (11) is an FDG model on $2n$ agents. In its digraph, which we call the *augmented graph*:

- a real agent i is influenced by a real agent j with weight $\lambda_i w_{ij}$;
- a real agent i is influenced by its virtual agent i' with weight $1 - \lambda_i$, so this edge exists only if agent i is prejudiced;
- a virtual agent i' is influenced only by itself, with weight 1, so its opinion never changes: it is stubborn (Definition 2.4).

Figure 4 shows the FJ model on the directed cycle of Figure 1(a) when agent 1 is the only prejudiced agent, and the corresponding augmented graph. In panel (a) the prejudice is drawn as an input entering agent 1, since it is not an agent of the network but an external piece of information; in panel (b) it becomes the virtual agent $1'$, and the edge follows our convention: it goes from 1 to $1'$, because agent 1 listens to $1'$. In social terms, a prejudice behaves as an additional agent that never changes its opinion: a prejudiced agent keeps listening to the others, and also listens to this stubborn agent placed beside it.



(a) FJ model on the directed cycle (b) augmented graph

Figure 4: The FJ model on the directed cycle of Figure 1(a) when agent 1 is the only prejudiced agent ($\lambda_1 < 1$, $\lambda_i = 1$ for $i \neq 1$), and its augmented graph. Unlabelled edges have weight 1. (a) The prejudice u_1 enters agent 1 as an external input, with weight $1 - \lambda_1$; this arrow is not an edge of G. (b) The input becomes the virtual agent $1'$ (dashed), which is stubborn and keeps the opinion u_1. The virtual agents $2', \dots, 6'$ are stubborn agents that nobody listens to, and are not drawn.

The augmented graph also shows, from the side of the FDG model, why consensus is not the right notion for the FJ model. This observation is ours. The augmented model (11) is an FDG model, so the classification of Table 1 applies to it. Each virtual agent is influenced only by itself, so it is a sink of the condensation digraph of the augmented graph, and it is aperiodic because of its self-loop. Hence the augmented graph has at least n sinks and it is not strongly connected; for $n \geq 2$, the consensus cases (i) and (iii) of Table 1 never occur, and neither does case (ii). Only two outcomes remain.

- *Disagreement* (case (iv)), when all the sinks are aperiodic. If all agents are P-dependent, the only sinks are the n virtual agents, since every real agent is influenced, directly or through intermediaries, by some prejudiced agent j, and hence by j'. This is the situation of Remark 2.5: by Theorem 2.3, every real agent converges to a convex combination of the opinions of the stubborn virtual agents, i.e. of the prejudices, which is exactly (9). If some agents are P-independent, the sinks of C^{22} are further sinks of the augmented graph, and all of them are aperiodic if and only if W^{22} is regular.
- *No convergence* (case (v)), when at least one sink of C^{22} is periodic, i.e. when W^{22} is not regular.

These two outcomes are exactly the two alternatives of Theorem 3.2. The disagreement of case (iv) concerns all the $2n$ agents of the augmented model, whose virtual agents keep their different values u_i forever ($x'(0) = u$ is part of the initial condition); it does not imply that the final opinions of the real agents differ, which, as discussed above, depends on V and u.

3.1.4 Influence centrality

In the FDG model the social power w is defined only when the group reaches a consensus (Section 2.1.2), which is not the relevant notion for the FJ model (Section 3.1.3). Nor does the augmented FDG model (11) help: its graph has at least n sinks, so for $n \geq 2$ it never reaches a consensus and its social power is not defined (this remark is ours). A different measure is needed, and [5] discusses an extension of the social power to the FJ model, introduced by Friedkin.

Influence centrality. As in [5], we confine ourselves to the asymptotically stable FJ model with $u = x(0)$, the traditional assumption of the Friedkin–Johnsen theory (Section 3.1.1); then $x(\infty) = Vx(0)$ by (9). The assumption $u = x(0)$ is what gives the initial opinions a role: in general $x(\infty) = Vu$ does not depend on $x(0)$, so the initial opinions would have no weight in the final opinions. The social power of agent i is generalized as the mean weight of its initial opinion in determining the group members'

final opinions [5]. These mean weights are the elements of the non-negative vector c such that

$$c = n^{-1}V^\top \mathbf{1}_n, \qquad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i(\infty) = \frac{1}{n}\mathbf{1}_n^\top V x(0) = c^\top x(0), \tag{12}$$

and c_i is called the *influence centrality* of agent i [5]. In terms of V, c_i is the average of the i-th column of V, whose entries are the weights of $x_i(0)$ in the final opinions of the single agents. Since V is stochastic, c is non-negative and $c^\top \mathbf{1}_n = 1$, like w (Theorem 2.1), so the two measures can be compared directly. Similar to French's social power, the influence centrality is generated by an opinion formation mechanism [5].

The relation with the social power is made precise in [5]. The convergence and the consensus of the FDG model are equivalent, respectively, to the regularity and the full regularity of the stochastic matrix A [5] (Definition 1.10). For a fixed W, let $\Lambda = \alpha I_n$ with $\alpha \in (0, 1)$, i.e. the same susceptibility for all agents, and consider the corresponding matrix V_α and vector c_α. For $\alpha = 0$, i.e. $\Lambda = 0$, every agent keeps its initial opinion, $V = I$ and $c_0 = n^{-1}\mathbf{1}_n$: the social power is uniformly distributed between all agents [5].

Lemma 3.5 ([5])**.** *Let W be a fully regular matrix and w stand for the vector of French's social power, corresponding to the FDG model* (1) *with $A = W$. Then, $w = \lim_{\alpha\to 1-0} c_\alpha$.*

The limit cannot be replaced by $\alpha = 1$: for $\Lambda = I$ the FJ model is the FDG model, which is not asymptotically stable, and $I - W$ is singular since $1 \in \mathrm{spec}(W)$ (Theorem 1.7), so V is not defined. In social terms, the influence centrality moves from the uniform distribution, when every agent is anchored to its own initial opinion, to French's social power, when the weight of the prejudices vanishes.

3.2 Stubbornness and influence centrality in the FJ model

Unless stated otherwise, the examples of this section use as matrix W the two networks of Figure 1 without self-loops, the initial opinions $x(0) = (0,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 1)^\top$ of Section 2.2, and prejudices equal to the initial opinions, $u = x(0)$, as is traditional in the Friedkin–Johnsen theory (see [5]). This choice compares the two models on the same data, and it is required by the influence centrality of Section 3.1.4; each figure states its own exceptions.

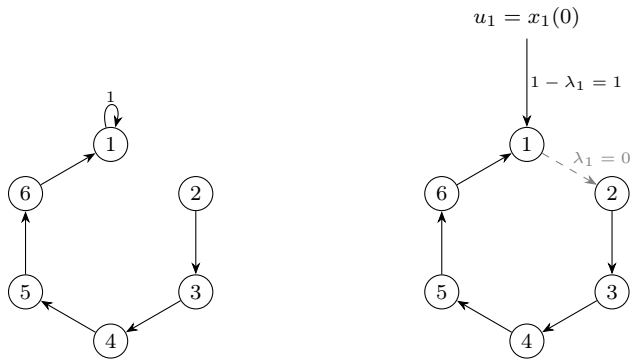
3.2.1 Modelling stubbornness: prejudices and stubborn agents

Stubborn agents in the two models. A stubborn agent of the FJ model is an agent with $\lambda_i = 0$ and $u_i = x_i(0)$ (Section 3.1.1). As in the FDG model, stubbornness is the extreme case of a graded mechanism, here the prejudice: an agent with $0 < \lambda_i < 1$ is prejudiced but not stubborn, since its opinion moves, even if [5] informally calls prejudiced agents "partially" stubborn. Unlike the FDG model, where $a_{ii} = 1$ suffices, a condition on the matrix only, here $\lambda_i = 0$ alone does not: stubbornness also requires $u_i = x_i(0)$, a condition on the input.

From one model to the other. The FJ model and the FDG model with stubborn agents can be translated into each other. In one direction, Section 3.1.3 has shown, following [5], that an FJ model is an FDG model on an augmented graph, in which every prejudice becomes a virtual stubborn agent (Figure 4). In the other direction, an FDG model with stubborn agents can be written as an FJ model in two ways: the rewriting of [5], which keeps each stubborn agent as an agent of the FJ model, and a rewriting of ours, which removes it and turns it into the prejudice of the agents that listen to it. The second one is the augmented graph read backwards. The two rewritings are presented in this order.

The stubborn agent as an input. Following [5], an FDG model with stubborn agents can be rewritten as the FJ model (4), with $\lambda_i = 0$ and $u_i = x_i(0)$ for every stubborn agent and $\lambda_i = 1$, $u_i = 0$ for the others (Figure 5). The two models produce the same trajectories: for a stubborn agent, $x_i(k+1) = x_i(k)$ in the FDG model and $x_i(k+1) = u_i = x_i(0)$ in the FJ model; every other agent has $\lambda_i = 1$ and the same row of the matrix in both models. The row of W of a stubborn agent plays no role, since it is multiplied by $\lambda_i = 0$; this is why the FJ model of Figure 6 can use the network

without the self-loop. The rewriting turns the stubborn agent from a state into an input: in the FJ model its opinion is the constant prejudice u_i. Note that in this rewriting the opinion of the stubborn agent appears twice: as its initial state $x_i(0)$ and as its prejudice $u_i = x_i(0)$.



(a) FDG model, $a_{11} = 1$ (b) FJ model, $\lambda_1 = 0$

Figure 5: (a) The FDG model on the directed cycle of Figure 1(a) with $t = 1$: agent 1 is stubborn and has only its self-loop. (b) The FJ model obtained from it with the rewriting of [5]: $\lambda_1 = 0$, $u_1 = x_1(0)$, $\lambda_i = 1$ for $i \neq 1$. Unlabelled edges have weight 1. In (b) the edge $1 \to 2$ is still an edge of G, but its weight in ΛW is $\lambda_1 = 0$ (dashed); the opinion of agent 1 enters as the external input $u_1 = x_1(0)$, with weight $1 - \lambda_1 = 1$, as the prejudice in Figure 4(a).

Figure 6 simulates the FDG model with $A(1)$ and its FJ rewriting on both networks. In the FJ model agent 1 is the only prejudiced agent, and its attachment $1 - \lambda_1$ takes the place of the self-weight t of Section 2.2: this choice of ours makes the two parameters directly comparable, both equal to 1 in the stubborn case.

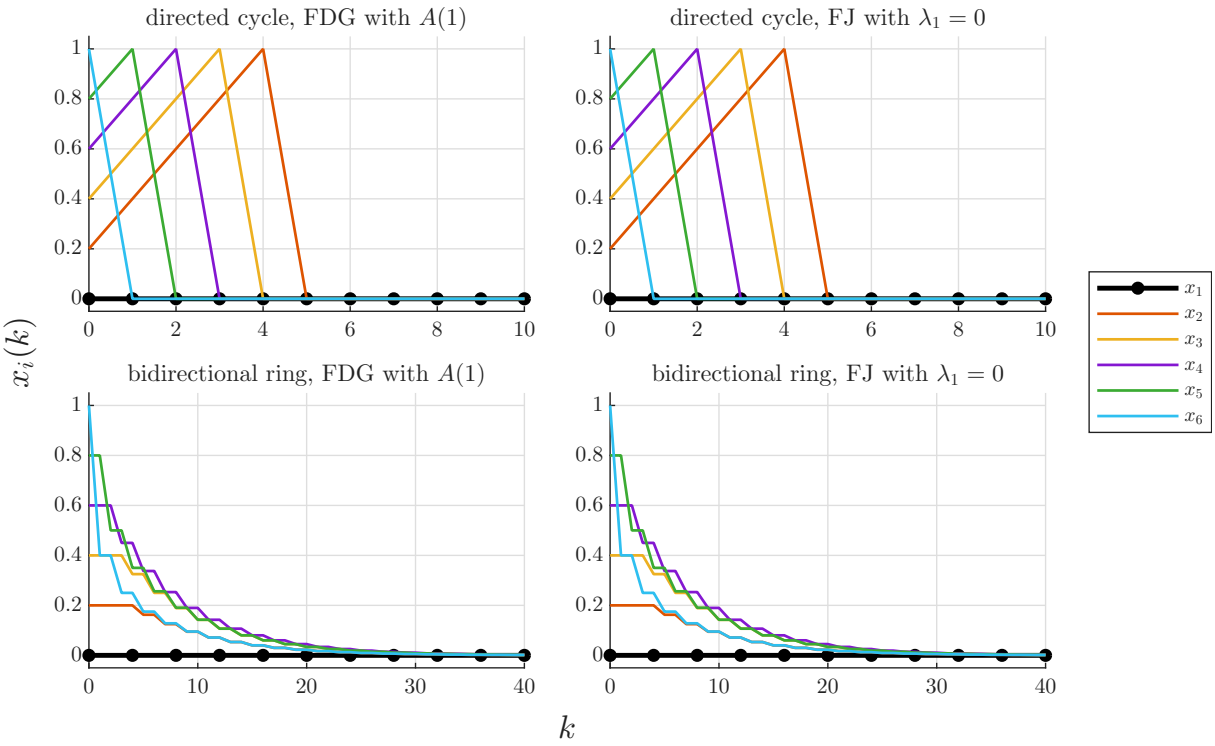


Figure 6: Opinions $x_i(k)$ in the FDG model with $A(1)$ (left) and in the FJ model with $\lambda_1 = 0$, $\lambda_i = 1$ for $i \neq 1$ and $u = x(0)$ (right), on the directed cycle ($K = 10$, top) and on the bidirectional ring ($K = 40$, bottom), with $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$. Agent 1 is black and marked with dots.

State or input. The two models of Figure 6 have the same trajectories, but not the same stability. The FDG model with stubborn agents converges under the following condition.

Corollary 3.6 ([5]). *Let the group have $s \geq 1$ stubborn agents, influencing all other individuals (i.e. every other node of G is connected by a walk to the set of stubborn nodes). Then the FDG model* (1) *is convergent.*

Under the assumption of Corollary 3.6, every agent of the FJ model obtained by the rewriting is prejudiced or influenced by a prejudiced agent, so the FJ model is asymptotically stable (Corollary 3.3), whereas the FDG model is convergent but not asymptotically stable, since a row-stochastic matrix always has the eigenvalue 1 (Theorem 1.7, see also [5]). The explanation given in [5] is that changing the initial condition $x(0)$ of the FJ model leaves the prejudice vector u unchanged, while in the FDG model the prejudice is part of the state vector.

Figure 7 starts the two models again from $x_1(0) = 0.5$, leaving u unchanged: now $u_1 = 0 \neq x_1(0)$, and agent 1 of the FJ model is prejudiced but no longer stubborn.

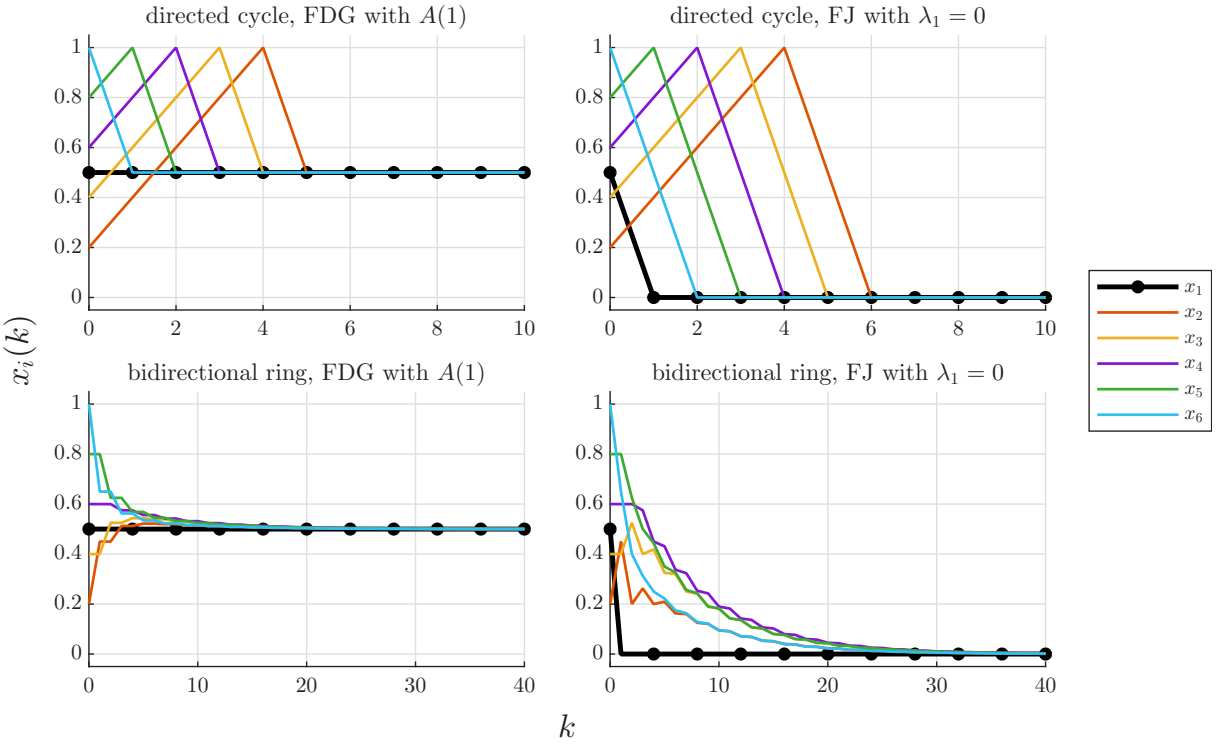


Figure 7: The models of Figure 6 started from $x_1(0) = 0.5$, with the other initial opinions and the prejudice vector $u = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$ unchanged, so $u_1 \neq x_1(0)$. Left: FDG with $A(1)$; right: FJ with $\lambda_1 = 0$. Top: directed cycle, $K = 10$; bottom: bidirectional ring, $K = 40$.

Figure 7 shows this difference. Moving $x_1(0)$ from 0 to 0.5 moves the stubborn agent of the FDG model, which keeps the new value forever, and the whole group follows it to 0.5. In the FJ model the same change does not touch $u_1 = 0$: agent 1 goes back to u_1 at the first step, and the group converges to the same limit Vu as before. In social terms, in the FDG model stubbornness is a memory inside the state, which is never forgotten; in the FJ model it is an external input, and the state forgets $x(0)$.

How the perturbation evolves. Let $x(k)$ and $\tilde{x}(k)$ be the trajectories of Figures 6 and 7, started from $x_1(0) = 0$ and $x_1(0) = 0.5$. Figure 8 shows, for each agent, their difference $\delta(k) = \tilde{x}(k) - x(k)$, with $\delta(0) = 0.5\, e_1$.

The difference $\delta(k)$ shows where the perturbation of $x_1(0)$ is at each step. In the FJ model, subtracting the two equations (4) written for $\tilde{x}$ and for x, the term $(I - \Lambda)u$ cancels, since the prejudices are the same, and (this computation is ours)

$$\delta(k+1) = \Lambda W\, \delta(k).$$

This is the same recursion as (10): both $\delta(k)$ and $e(k)$ are differences of two solutions with the same u, the second one being, for $e(k)$, the constant solution $x(\infty)$. The two quantities, however, measure

different things: $e(k)$ tells how far a single trajectory still is from its limit, $\delta(k)$ how much of the perturbation is still present.

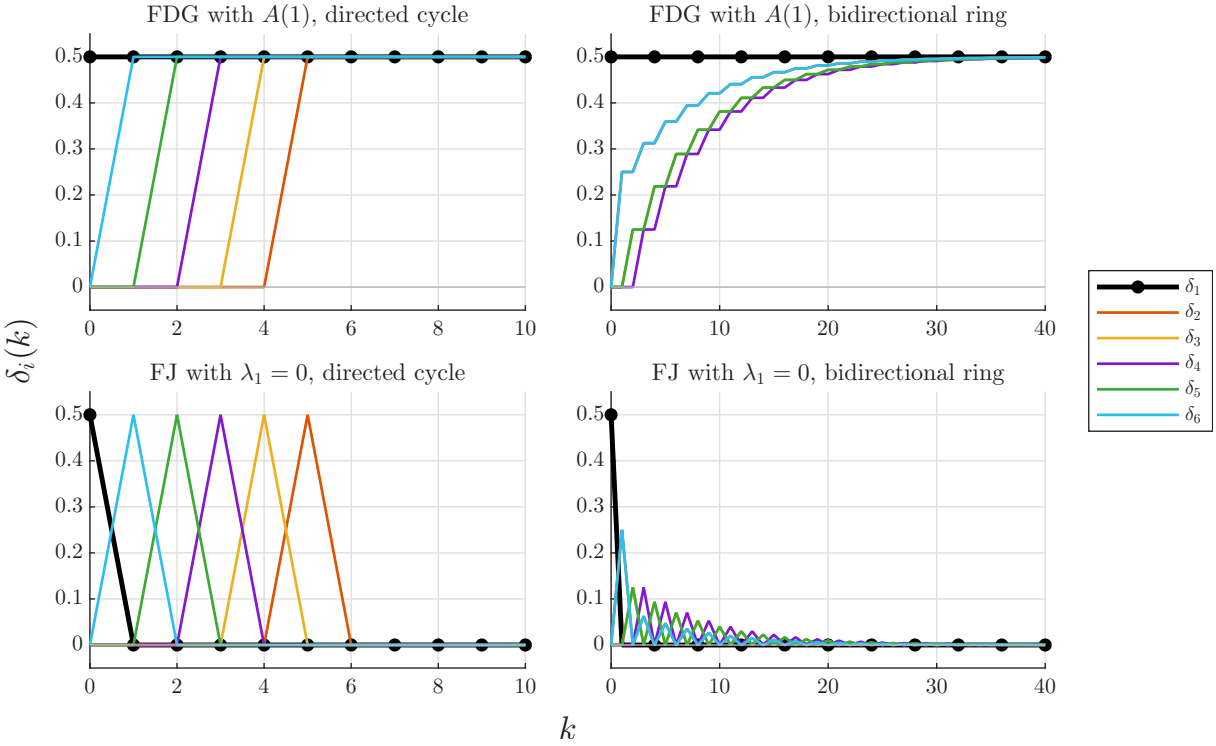


Figure 8: Difference $\delta_i(k) = \tilde{x}_i(k) - x_i(k)$ between the trajectories of Figure 7 and those of Figure 6, i.e. the effect of changing $x_1(0)$ from 0 to 0.5. Top: FDG with $A(1)$; bottom: FJ with $\lambda_1 = 0$. Left: directed cycle, $K = 10$; right: bidirectional ring, $K = 40$. The thin grey line marks $\delta_i = 0$.

Since all agents are P-dependent, ΛW is Schur stable (Corollary 3.3) and $\delta(k) \to 0$: the perturbation is forgotten. On the directed cycle it travels backwards along the cycle, from agent 1 to agents $6, 5, \ldots, 2$, and when it comes back to agent 1 it is cancelled, since $\lambda_1 = 0$: $\delta(k) = 0$ for $k \geq 6$. On the bidirectional ring it is split between the neighbours and fades gradually. In the FDG model there is no input to cancel, and the same computation gives $\delta(k+1) = A\,\delta(k)$ (ours). Agent 1 is the only globally reachable node, with social power $w = e_1$, so by Theorem 2.1 $\delta(k) \to (w^\top \delta(0))\mathbf{1}_6 = 0.5\,\mathbf{1}_6$: the stubborn agent keeps the perturbation and passes it to everybody.

The stubborn agent as a prejudice of its neighbours. The rewriting of [5] keeps the stubborn agent as an agent of the FJ model. A different rewriting, which is ours, removes it. Consider the bidirectional ring with $t = 1$ (Figure 9(a)): agent 1 is stubborn, and agents 2 and 6 listen to it with weight $1/2$. Since $x_1(k) = x_1(0)$ for all k, the update of agent 6 is

$$x_6(k+1) = \tfrac{1}{2}\,x_5(k) + \tfrac{1}{2}\,x_1(k) = \tfrac{1}{2}\,x_5(k) + \tfrac{1}{2}\,x_1(0),$$

which is the update (4) of an agent with $\lambda_6 = 1/2$, prejudice $u_6 = x_1(0)$ and influence weight 1 on agent 5; the same holds for agent 2. Hence agents $2, \ldots, 6$ obey an FJ model on five agents (Figure 9(b)), with $\lambda_2 = \lambda_6 = 1/2$, $u_2 = u_6 = x_1(0)$ and $\lambda_3 = \lambda_4 = \lambda_5 = 1$, whose trajectories coincide with those of the FDG model (bottom-left panel of Figure 6): the stubborn agent has become the common prejudice of the agents that listen to it. Conversely, the augmented graph of this FJ model (Section 3.1.3) has two virtual agents $2'$ and $6'$, both with opinion $x_1(0)$; merging them into a single node gives back the FDG model, in which the stubborn agent 1 plays the role of a virtual agent.



(a) FDG model, agent 1 stubborn

(b) FJ model on agents $2, \dots, 6$

Figure 9: (a) The FDG model on the bidirectional ring of Figure 1(b) with $t = 1$: agent 1 is stubborn; unlabelled edges have weight 1/2. (b) The FJ model on agents $2, \dots, 6$ obtained by removing agent 1 (our rewriting): $\lambda_2 = \lambda_6 = 1/2$, $u_2 = u_6 = x_1(0)$, $\lambda_3 = \lambda_4 = \lambda_5 = 1$. The opinion $x_1(0)$ enters agents 2 and 6 as their common prejudice, with weight $1 - \lambda_2 = 1 - \lambda_6 = 1/2$; every edge between agents has weight 1/2.

The two rewritings differ in one point. In both of them the FDG model is not asymptotically stable while the FJ model is (all its agents are P-dependent, Corollary 3.3), because the opinion of the stubborn agent is a state in the first and an input in the second. In the rewriting of [5], however, this opinion appears twice, and moving the initial state of the stubborn agent alone separates the two models (Figure 7). In our rewriting it appears only once, as the prejudice $u_2 = u_6 = x_1(0)$: there is no initial condition of the FJ model that corresponds to moving the stubborn agent alone, and the two models have the same trajectories for every initial condition.

3.2.2 Prejudices, stubbornness and convergence

Figure 10 is the FJ counterpart of Figure 2: agent 1 is the only prejudiced agent, and its attachment $1 - \lambda_1$ takes the three values 0, 1/2 and 1 of the self-weight t. In the middle column the grey dashed line marks the consensus value ≈ 0.4286 reached by the FDG model with $t = 1/2$ (Section 2.2.3), so that the two models can be compared at the same value of the parameter.

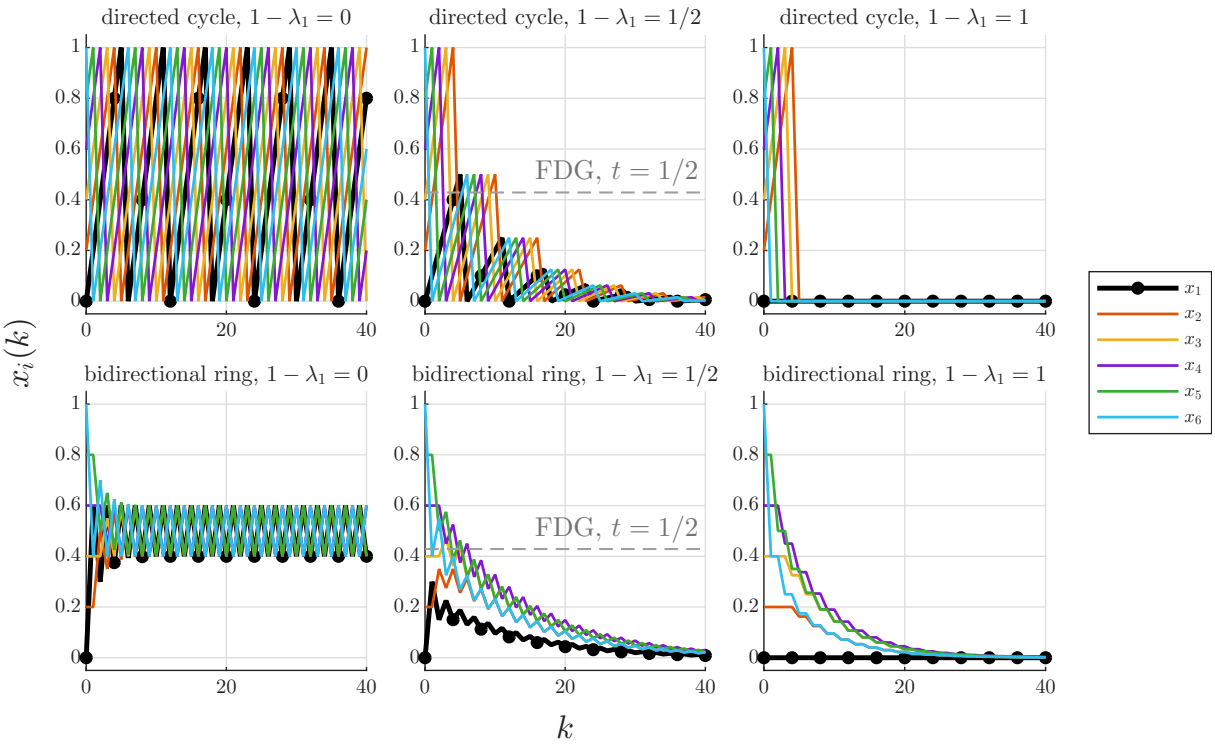


Figure 10: Opinions $x_i(k)$ in the FJ model with a single prejudiced agent, agent 1, for three values of its attachment $1-\lambda_1$ (columns), with $\lambda_i=1$ for $i\neq 1$, $u=x(0)=(0,\,0.2,\,0.4,\,0.6,\,0.8,\,1)^\top$ and $K=40$. Top: directed cycle; bottom: bidirectional ring. In the middle column the grey dashed line marks the consensus value of the FDG model with $t=1/2$, ≈ 0.4286.

In Figure 11, instead, every agent is prejudiced, with the same susceptibility; the triangles mark, agent by agent, the limit Vu.

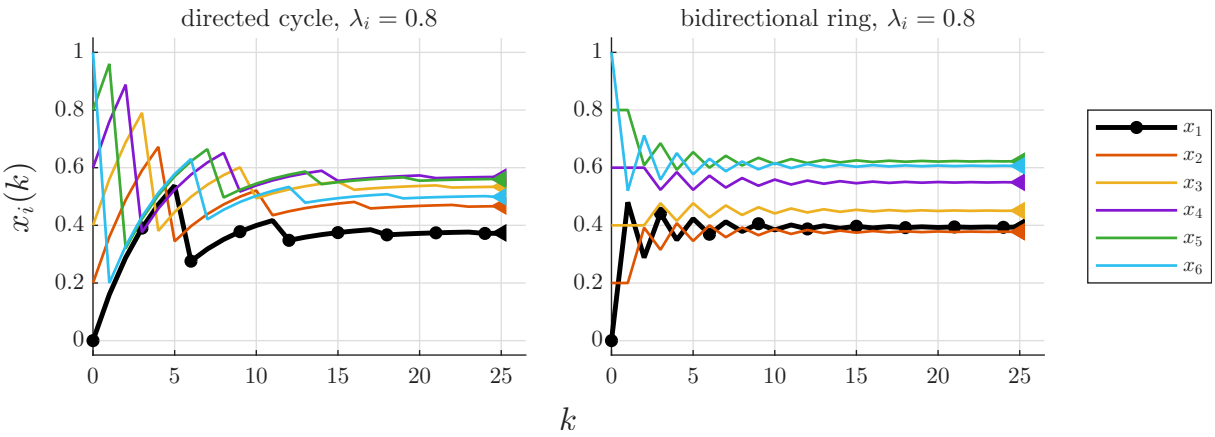


Figure 11: Opinions $x_i(k)$ in the FJ model with every agent prejudiced, $\lambda_i=0.8$ for all i, $u=x(0)=(0,\,0.2,\,0.4,\,0.6,\,0.8,\,1)^\top$ and $K=25$, on the directed cycle (left) and on the bidirectional ring (right). The triangles mark the limit Vu of (9).

The two networks of Figure 1 are strongly connected, so a single prejudiced agent is enough to reach every other agent. To show what happens when this is not the case, Figure 12 introduces an auxiliary network on $n=6$ agents: agents 1, 2 and 3 form a closed cycle, and agents 4, 5 and 6 form a chain that feeds into it. The dashed edge $2\to 1$ is an optional modification: when it is present, agent 2 splits its weight equally between agents 1 and 3.

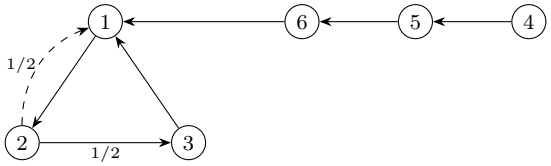


Figure 12: The auxiliary network used in Figure 13. The solid edges have weight 1; when the dashed edge $2 \to 1$ is present, the two edges leaving agent 2 have weight $1/2$ each.

Figure 13 uses this network with a single prejudiced agent, in four configurations: (a) without the dashed edge, agent 2 prejudiced; (b) without the dashed edge, agent 5 prejudiced; (c) as (b), with the dashed edge $2 \to 1$; (d) as (c), with $x_2(0) = 1$, hence $u_2 = 1$.

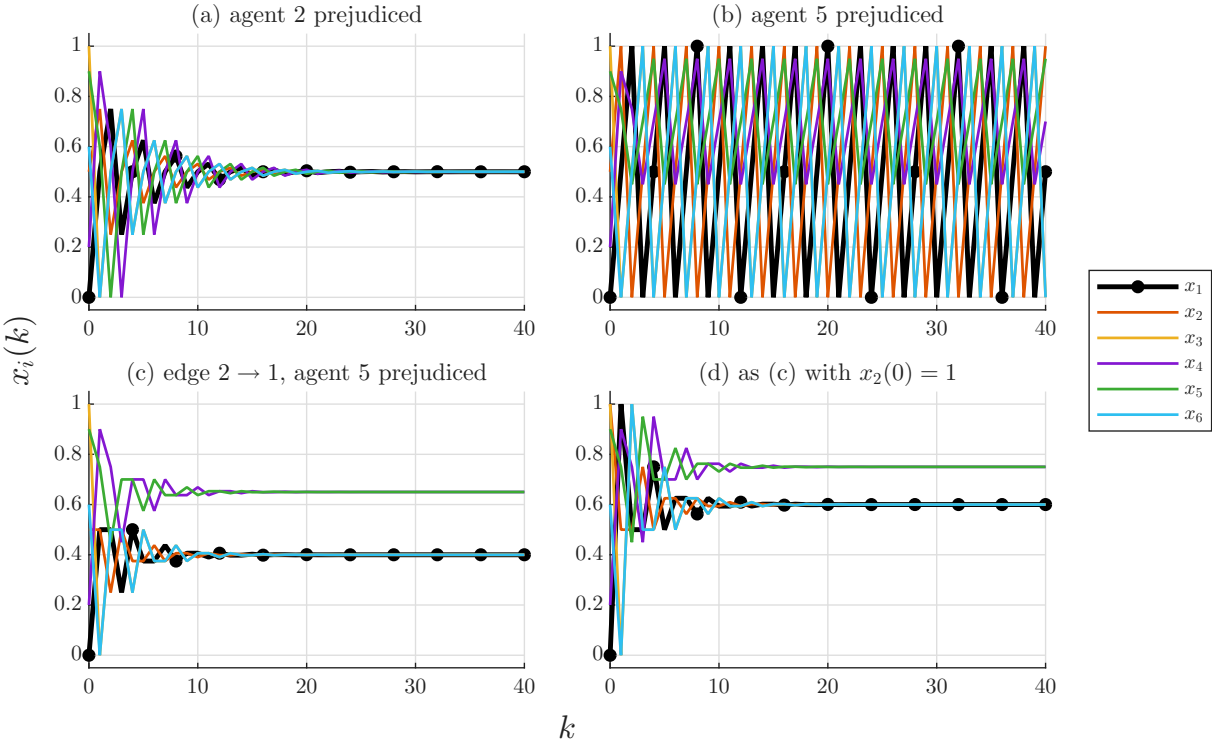


Figure 13: Opinions $x_i(k)$ in the FJ model on the auxiliary network of Figure 12, with $x(0) = (0, 0.5, 1, 0.2, 0.9, 0.6)^\top$, $u = x(0)$, $K = 40$, the prejudiced agent having $\lambda = 0.5$ and every other agent $\lambda_i = 1$. (a) agent 2 prejudiced; (b) agent 5 prejudiced; (c) as (b) with the edge $2 \to 1$ added; (d) as (c) with $x_2(0) = 1$.

3.2.3 Prejudices, stubbornness and influence centrality

Figure 14 is the FJ counterpart of Figure 3: the influence centrality c_1 of agent 1 and the mean final opinion $c^\top x(0)$ of (12), as functions of its attachment $1 - \lambda_1$, with the other agents prejudiced ($\lambda_i = 0.8$); the FDG social power $w_1(t)$ and consensus value $w(t)^\top x(0)$, with the self-weight t on the same axis, are drawn as a reference.

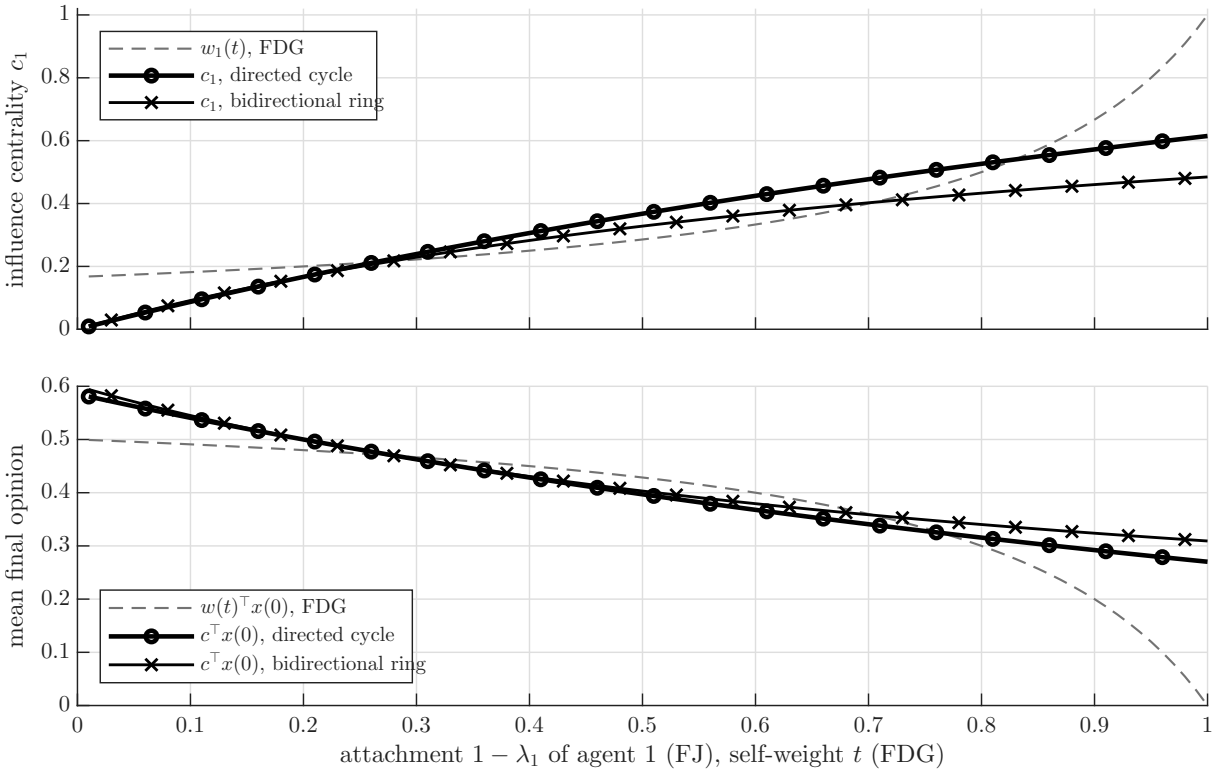


Figure 14: Top: influence centrality c_1 of agent 1 in the FJ model, as a function of its attachment $1-\lambda_1 \in (0,1]$, with $\lambda_i = 0.8$ for $i \neq 1$ and $u = x(0) = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$, on the directed cycle (dots) and on the bidirectional ring (crosses); the grey dashed line is the social power $w_1(t)$ of the FDG model of Figure 3. Bottom: the corresponding mean final opinion $c^\top x(0)$, with the FDG consensus value $w(t)^\top x(0)$ as a grey dashed reference.

Figure 15 illustrates Lemma 3.5. Since the lemma requires W fully regular, here $W = A(1/2)$, the network of Figure 1 with the self-weight $t = 1/2$, and the susceptibility is the same for every agent, $\Lambda = \alpha I$; the figure compares $c_1(\alpha)$ and $c_2(\alpha)$ with the social power w_1 and w_2 of the FDG model with the same matrix.

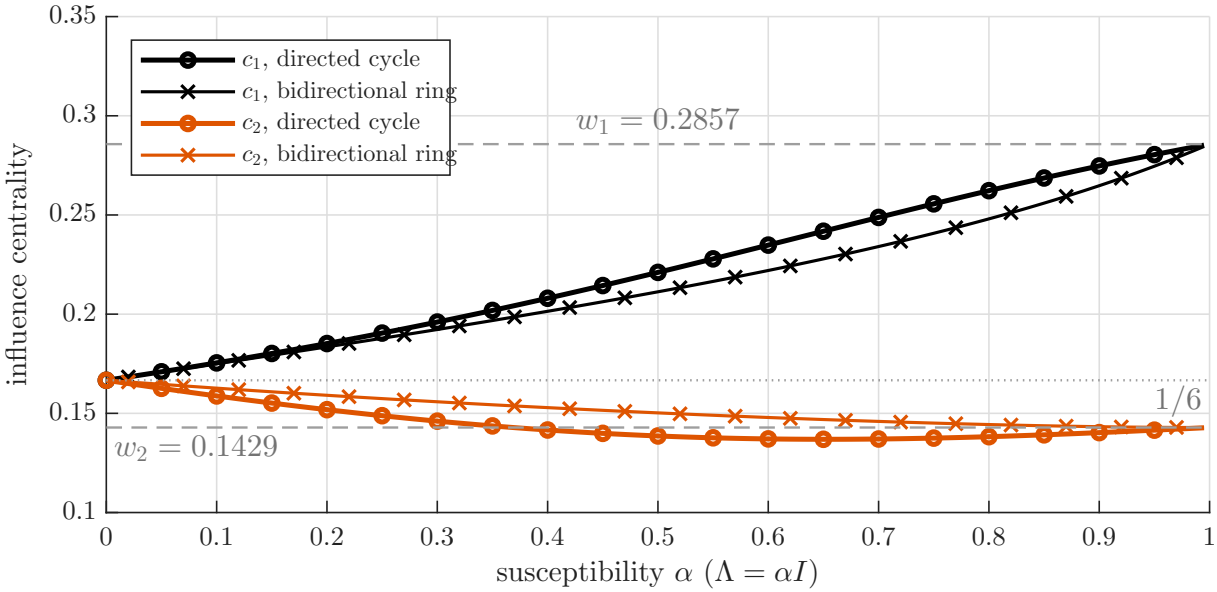


Figure 15: Influence centrality c_1 and c_2 in the FJ model with $W = A(1/2)$ and $\Lambda = \alpha I$, $\alpha \in [0, 0.995]$, and $u = x(0) = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$, on the directed cycle (dots) and on the bidirectional ring (crosses). The dashed lines mark the social power w_1 and w_2 of the FDG model with the same matrix $A(1/2)$, the dotted line the uniform value $1/6$.

4 Conclusions

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